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12.4 DIFFERENTIAL CALCULUS

12.4.1 Meaning

The fundamental operation of calculus is differentiation. Derivative is used to express the rate of change in any function. Derivative means a change in the dependent variable with respect to small change (closer to zero) in independent variable.

Let the function be,

$y = f(x)$

Differentiating y with respect to x is,

$$\frac{d(y)}{dx} = \frac{df(x)}{dx}$$

12.4.2 Some Standard Forms of Differentiation

(Constant, addition and subtraction only)

1. $\frac{d(c)}{dx} = 0$ where C is a constant.

   (Read differentiation of 'C' with respect to 'x' is)

2. $\frac{d(x^n)}{dx} = nx^{n-1}$

3. $\frac{d(x)}{dx} = 1x^{1-1} = 1x^0 = 1$

4. $\frac{d(u+v)}{dx} = \frac{du}{dx} + \frac{dv}{dx}$

5. $\frac{d(u-v)}{dx} = \frac{du}{dx} - \frac{dv}{dx}$

Key Note

(Any non-zero real number)^0 = 1

Example: 12.12

If $Y = 4$, then find $\frac{dy}{dx}$

Solution:

$Y = 4$, here 4 is a constant.Differentiation of constant function is zero.

So, $\frac{dy}{dx} = \frac{d(4)}{dx} = 0$

Example: 12.13

Find the slope of the function $y = 6x^3$ for any value of x.

Solution:

Given $y = 6x^3$

Slope $= \frac{dy}{dx}$

$\frac{dy}{dx} = 6(3)x^{3-1} = 18x^2$ for any value of x.

Example: 12.14

What is the slope of the function $y = 5x^4$ when $x = 10$?

Solution:

Given function $y = 5x^4$

Slope $= \frac{dy}{dx}$

$\frac{dy}{dx} = 5(4)x^{4-1}$

$= 20x^3$

When $x = 10$, then slope $= 20\ (10)^3$

$= 20{,}000$,

Therefore Slope is 20,000.